

Interaction & Embodiment

Let's begin the discussion by answering a few questions.

Interaction & Embodiment Quiz

An AI system produces a highly confident recommendation, but the system operates in an environment where an incorrect decision could have significant consequences.

What does HITL recognize about AI confidence?

A

High confidence guarantees that the recommendation is correct

B

Human review is unnecessary when confidence is high

C

The AI should be retrained whenever confidence is high

D

Model confidence alone should not determine whether a high-risk action is executed

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Confidence $\neq$ Authorization

HITL provides an external control point for consequential decisions.

Interaction & Embodiment Quiz

**An AI workflow has three types of actions:
low-risk actions, medium-risk actions, and high-risk actions.**

Which approach best represents a well-designed HITL system?

A

Require human approval for all three types of actions

B

Allow the AI to execute all three types autonomously

C

Automatically execute low-risk actions and introduce human review for higher-risk actions

D

Allow the AI to decide whether human approval is necessary after execution

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Interaction & Embodiment Quiz

An AI agent reaches a decision point where human approval is required. The workflow should stop, allow a person to review the proposed action, and then continue from the same point after approval.

Which workflow pattern best represents this HITL process?

A

Generate → Execute → Ask for approval

B

Generate → Execute → Undo if the human disagrees

C

Generate → Pause → Human Review → Resume → Execute

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AI-Generated SQL in Healthcare

Why autonomous database execution needs governance

RISK

LLMs may generate incorrect, ambiguous, or unsafe SQL.

Healthcare adds privacy, compliance, and operational sensitivity.

FAILURE MODES

- SQL hallucinations
- Unintended writes
- Data exposure
- Destructive operations

GUARDRAIL

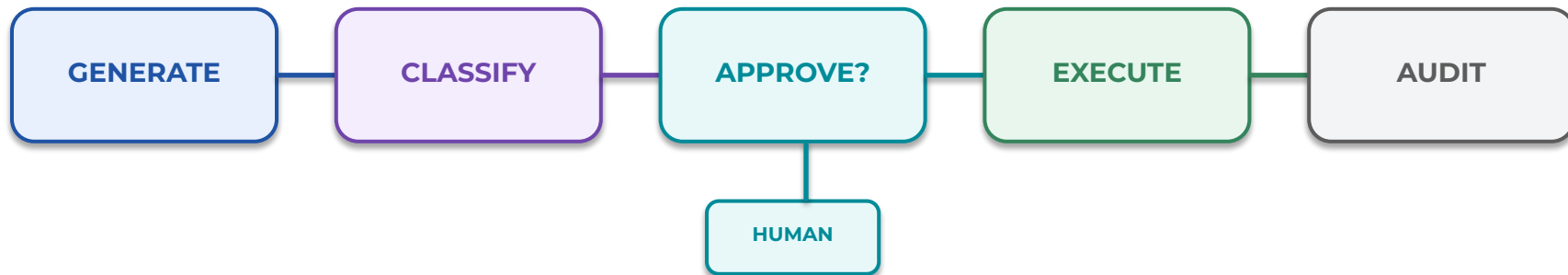
Separate generation from validation, risk evaluation, execution, and audit.

Automate low-risk work; control high-risk work.

KEY IDEA : An LLM can generate SQL; it should not be the final authority on what executes.

HITL (Human-in-the-Loop)

A practical safety boundary for sensitive AI operations

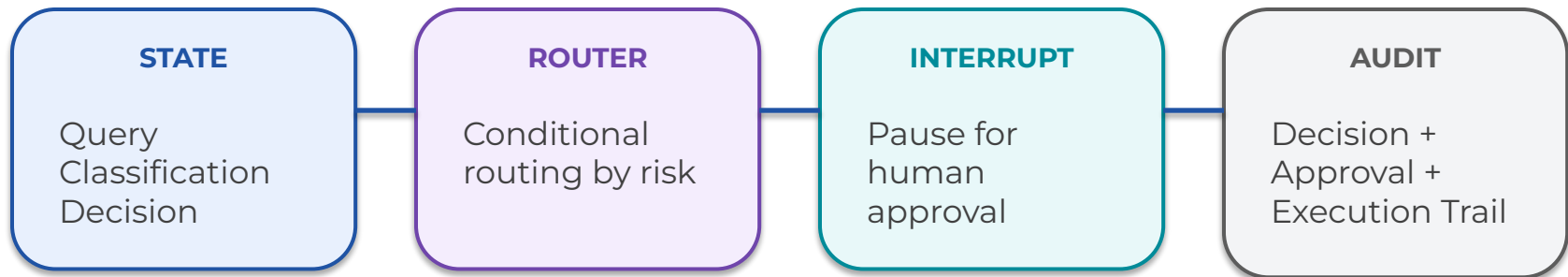


Human review is triggered by risk - not necessarily by every query.

KEY IDEA : HITL balances automation with human oversight and accountability.

LangGraph Workflow

State + routing + interrupts for governed execution



STATE PERSISTENCE → pause → human decision → resume with context

KEY IDEA : Interrupts create the approval gate; persistence makes the workflow resumable.

SQL Risk Classification

Three tiers determine the execution path

READ

SELECT / read-only

Auto-execute under
safeguards

WRITE

INSERT / UPDATE /
DELETE

Route through
stronger controls

UNSAFE

Destructive /
prohibited /
suspicious operations

Reject or require
explicit review

Risk level → routing decision → execution / approval / rejection

KEY IDEA : Classification drives routing; model confidence alone should not.

Recap

Five ideas to carry forward

01

LLM ≠ authority

Generate SQL, but govern what executes.

02

HITL

Use human oversight for sensitive actions.

03

Classify

READ / WRITE / UNSAFE drives routing.

04

Interrupt

Pause, approve, and resume safely.

05

Audit

Record decisions and execution for accountability.

KEY IDEA : Safe AI = useful automation + explicit controls + auditability.

Q&A



Power Ahead !

